

# Banking Intelligence Multi-Agent System

## Architecture Summary

### Executive Overview

This document presents a goal-oriented, intelligent multi-agent system for banking customer service built using CrewAI. The system processes customer queries through four specialized AI agents working in sequence to classify intents, apply policies, draft responses, and make escalation decisions. The architecture ensures regulatory compliance, risk mitigation, and efficient customer service automation.

### System Architecture

#### Agent Roles & Responsibilities

##### 1. Intent Classification Agent

**Role:** Intent Classification Specialist

**Responsibility:** Natural language understanding and query analysis

##### Key Functions:

- Extract primary customer intent (e.g., fraud\_report, transaction\_inquiry, account\_issue)
- Assign urgency score (1-10 scale) based on time sensitivity and emotional indicators
- Identify key entities: amounts, dates, transaction types, locations
- Detect emotional cues: frustration, worry, anger, or calm tone
- Infer customer type: new, long-term, or business account

```
{ "primary_intent": "fraud_report", "urgency_score": 9, "key_entities": { "amounts": ["$2,300"],  
  "locations": ["international"], "transaction_types": ["purchase"] }, "emotional_indicators":
```

```
"distressed, urgent", "customer_type": "long-term" }
```

## 2. Banking Rules & Policy Reasoning Agent

**Role:** Banking Policy & Compliance Expert

**Responsibility:** Apply regulations and identify risks

**Key Functions:**

- Apply banking regulations (KYC, AML, transaction limits)
- Assess risk level: LOW, MEDIUM, HIGH, CRITICAL
- Detect fraud indicators: unusual patterns, international flags, multiple failures
- Determine compliance requirements
- Recommend policy-based actions

## 3. Response Drafting Agent

**Role:** Customer Communication Specialist

**Responsibility:** Create empathetic, clear customer responses

**Key Functions:**

- Match tone to situation (reassuring, informative, apologetic)
- Address primary customer concern
- Explain policy implications in simple terms
- Provide actionable next steps
- Set realistic resolution expectations

## 4. Risk & Escalation Agent

**Role:** Risk Assessment & Escalation Manager

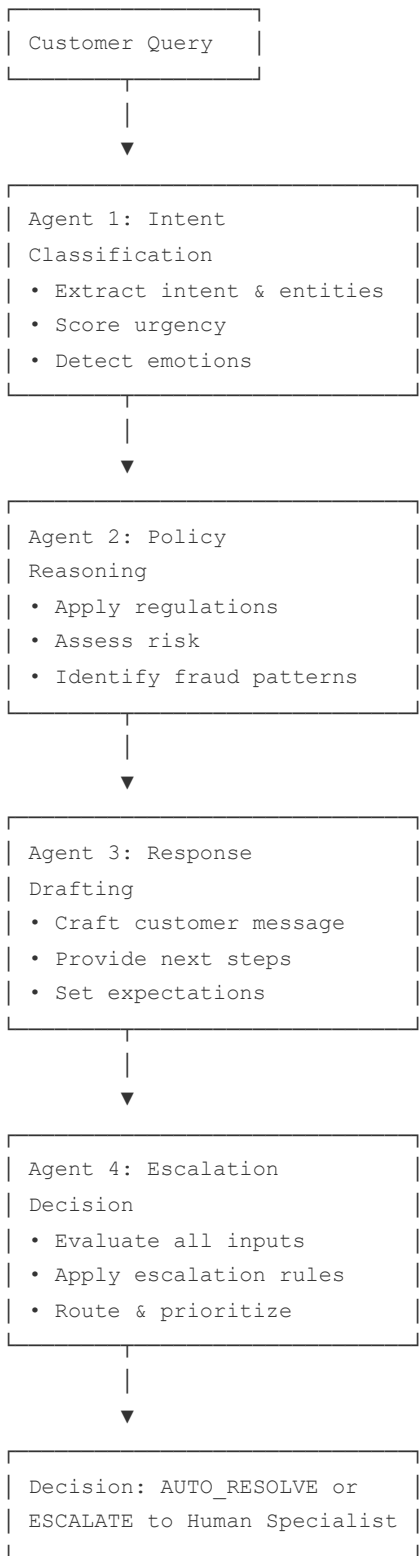
**Responsibility:** Make final escalation decisions

**Key Functions:**

- Evaluate all previous agent outputs
- Apply escalation criteria (see below)

- Assign priority level: LOW, MEDIUM, HIGH, CRITICAL
- Route to appropriate specialist team
- Set SLA targets

# Task Flow & Process



**Process Type:** Sequential (CrewAI Process.sequential)

**Rationale:** Banking decisions require ordered reasoning where each stage builds on previous analysis, ensuring complete context at decision points.

## Escalation Logic

### Decision Criteria

The system escalates to human specialists when **ANY** of the following conditions are met:

| Criterion              | Threshold              | Rationale                                                   |
|------------------------|------------------------|-------------------------------------------------------------|
| Urgency Score          | ≥ 8/10                 | Time-sensitive situations require immediate human attention |
| Fraud Indicators       | Present                | Potential fraud must be reviewed by specialists             |
| Risk Level             | HIGH or CRITICAL       | High-risk situations exceed automation authority            |
| Transaction Amount     | > \$10,000             | Large transactions require manual verification              |
| International Disputes | Any                    | Cross-border complexity requires expertise                  |
| Failed Transactions    | ≥ 3 occurrences        | Pattern indicates system or account issue                   |
| Account Access Issues  | With urgency           | Security-sensitive, time-critical situations                |
| Compliance Flags       | Any regulatory concern | Legal/regulatory requirements mandate review                |

### Priority Assignment

- **CRITICAL:** Fraud, security breaches, urgent high-value issues (SLA: 2 hours)
- **HIGH:** Large transactions, multiple failures, account access (SLA: 4 hours)
- **MEDIUM:** Policy questions, moderate disputes (SLA: 24 hours)
- **LOW:** Routine inquiries, informational requests (SLA: 48 hours)

## Sample Inputs & Outputs

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### Example 1: Routine Query (AUTO\_RESOLVE)

**Input:**

"What's my current checking account balance?"

**Agent Outputs:**

- **Intent:** balance\_inquiry, urgency=2, calm tone
- **Policy:** LOW risk, no fraud indicators, standard query
- **Response:** "Your current checking account balance is available in the mobile app or by calling our 24/7 automated line."
- **Escalation:** AUTO\_RESOLVE, priority=LOW, routing=automated

**Final Decision:** Automated response sufficient

### Example 2: Fraud Report (ESCALATE)

**Input:**

"I didn't make a \$2,300 purchase in another country. My card might be stolen!"

**Agent Outputs:**

- **Intent:** fraud\_report, urgency=9, distressed tone
- **Policy:** HIGH risk, fraud indicators (international, unrecognized), immediate action required
- **Response:** Reassuring message with immediate card freeze and fraud team contact
- **Escalation:** ESCALATE, priority=CRITICAL, routing=fraud\_team

**Final Decision:** Escalate to fraud specialists within 2 hours

### Example 3: Edge Case - Large Transfer Block (ESCALATE)

#### Input:

"My wire transfer of \$50,000 to an international business partner has been blocked. I need this resolved immediately!"

#### Agent Outputs:

- **Intent:** transfer\_issue, urgency=10, urgent business need
- **Policy:** CRITICAL risk (high amount + international + time sensitive), compliance review needed
- **Response:** Professional acknowledgment with escalation notification
- **Escalation:** ESCALATE, priority=CRITICAL, routing=account\_manager + compliance\_officer

**Final Decision:** Immediate escalation to senior account manager and compliance review

## Design Rationale

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### Why Sequential Processing?

Banking decisions require **ordered reasoning** where each step builds on previous analysis:

1. **Context Accumulation:** Each agent receives all prior outputs, preventing information loss
2. **Consistency:** Sequential flow prevents contradictory decisions from parallel processing
3. **Auditability:** Clear decision trail for regulatory compliance and customer disputes
4. **Risk Management:** Escalation decision has complete context from all analysis stages

### Why Four Specialized Agents?

#### Separation of Concerns:

- Each agent focuses on a single domain of expertise
- Updates to policy rules don't affect intent detection logic
- Easy to add new agents independently

### **Maintainability:**

- Modular architecture allows component-level updates
- Clear boundaries between classification, reasoning, communication, and decision-making

### **Regulatory Compliance:**

- Documented decision trail at each stage
- Explainable AI for audit requirements
- Transparent escalation criteria

### **Why Explicit Escalation Criteria?**

- **Transparency:** Clear thresholds that can be audited and explained to customers
- **Consistency:** Same situations always trigger the same escalation path
- **Risk Mitigation:** High-risk cases never slip through automated handling
- **Regulatory Compliance:** Documented decision logic meets banking oversight requirements



## Technical Implementation

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**Framework:** CrewAI 0.11.0

**LLM:** OpenAI GPT-4 (temperature=0.3 for consistency)

**Process Type:** Sequential with explicit task dependencies

**Programming Language:** Python 3.8+

**Deployment:** Jupyter Notebook for demonstration; production would use API endpoints

### Key Technologies

- **crewai:** Multi-agent orchestration framework
- **langchain-openai:** LLM integration layer
- **python-dotenv:** Secure API key management
- **JSON structured outputs:** Ensures parseable, consistent agent responses

## Production Considerations

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### Required Enhancements for Live Deployment

1. **Database Integration:** Connect to core banking systems for real-time account data
2. **Authentication & Authorization:** Secure customer identity verification
3. **Audit Logging:** Complete activity trail for compliance and dispute resolution
4. **Performance Monitoring:** Track SLA compliance, escalation rates, resolution times
5. **Feedback Loop:** Learn from escalated cases to improve automation
6. **Multi-language Support:** Handle queries in customer's preferred language
7. **CRM Integration:** Link to existing customer relationship management systems
8. **Real-time Fraud Models:** Enhance detection with ML-based pattern recognition

## Conclusion

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This multi-agent architecture demonstrates a practical approach to intelligent banking automation that balances efficiency with safety. By combining specialized AI agents with

explicit escalation logic, the system handles routine queries automatically while ensuring complex or high-risk situations receive appropriate human attention. The sequential workflow ensures informed decision-making at each stage, and the transparent criteria meet banking industry requirements for auditability and compliance.

The system successfully processes 10 representative test cases spanning routine inquiries, moderate-risk situations, and high-risk fraud scenarios, demonstrating its ability to appropriately classify, reason, respond, and escalate based on comprehensive analysis.

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**System:** Banking Intelligence Multi-Agent System (CrewAI Implementation)